

Calazans Macchiutti

Senior Data Scientist & AI / ML Engineer - Ph.D. in Physics

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Summary

Senior Data Scientist and AI/ML Engineer with a **Ph.D. in Physics**, combining a scientist's rigor with an engineer's delivery. Around four years of professional data science on top of a research career modeling large, complex, noisy datasets. I own problems end to end, from hypothesis and experimentation to **production deployment, monitoring and evaluation**, across three areas: **Generative AI** (LLM-powered agents, RAG, evals), **Machine Learning** (forecasting, anomaly and fraud detection, statistical modeling, computer vision), and **production engineering** (Python services, containers, cloud). I define the metric that matters, measure honestly, isolate what drives it, and ship what works.

Technical Stack

- **Generative AI & LLMs:** LLMs (Claude, OpenAI), **Retrieval-Augmented Generation (RAG)**, AI agents, agentic workflows, tool calling, prompt engineering, **structured output** (Pydantic), **LLM-as-judge / evaluation harnesses**, human-in-the-loop, LLM observability (cost, latency, quality), vendor-agnostic LLM clients (Claude/Ollama fallback); portable to LangChain / LangGraph and AWS Bedrock; MCP (Model Context Protocol)-aware
 - **Embeddings & Vector Search:** embeddings, semantic search, **vector databases** (Qdrant, FAISS, ChromaDB), SentenceTransformers (MiniLM)
 - **Machine Learning:** supervised and unsupervised learning, classification, regression, clustering, **gradient boosting** (XGBoost, LightGBM), **deep learning** (TensorFlow, Keras, PyTorch), **time series / forecasting** (ARIMAX, LSTM), **anomaly detection, fraud detection (KYC/KYT)**, recommendation basics, model evaluation (Average Precision, validation)
 - **Statistics & Modeling:** statistical and **probabilistic modeling**, hypothesis testing, experimentation, graph analysis, inference on noisy data, explainability (**SHAP**)
 - **Computer Vision:** object classification, image-based inspection
 - **NLP:** text classification, information extraction, RAG over documents
 - **MLOps & Production:** end-to-end **deployment and monitoring**, model serving, CI/CD, **Docker** (multi-stage, docker-compose), **Kubernetes** (self-healing), Git, **MLflow**; observability and reproducibility; **PySpark, Databricks**, distributed processing; **Google Cloud**
 - **Cloud:** **AWS** (S3, SageMaker, EC2, Lambda), **Microsoft Azure** (Functions, Azure ML)
 - **Programming & Data:** **Python** (Pandas, NumPy, SciPy, Polars), **SQL**, PostgreSQL, ETL, data pipelines, feature engineering, exploratory data analysis; APIs with **FastAPI**
 - **Data Viz & BI:** Power BI, Plotly, Streamlit, KPI frameworks and dashboards
 - **AI-assisted development:** GitHub Copilot, Cursor, Claude Code / ChatGPT
 - **Ways of working:** hypothesis-driven and iterative, remote/async, stakeholder communication (technical and executive), English and Portuguese
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Professional Experience

Senior Data Scientist / AI Engineer - ESPM

Aug 2024 - Present · Remote, Brazil

Stack: Python, SQL, RAG, embeddings, Qdrant, FAISS, ChromaDB, SentenceTransformers, OpenAI/Claude, AI agents, LLM-as-judge, structured output (Pydantic), FastAPI, Docker, Kubernetes, Azure Functions, Azure ML, Power BI, MLOps (deploy/monitoring)

- Designed and shipped an internal **Generative AI platform** for exam curation and generation: an **agentic pipeline with RAG** (vector retrieval over embeddings) on top of LLMs (Claude), with two LLM judgment points using **structured output**, human-in-the-loop, and eval-driven validation against a baseline.
- Kept the LLM **out of the critical selection loop** - final assembly is deterministic and constraint-based - guaranteeing reproducibility and real, traceable IDs while using the model where it adds value.
- Made cost and quality **observable before deploy**: per-question LLM call budget, tracing of both judgment points, and continuous evaluation against a baseline, with a human review queue where the judges disagree.
- Built the **data and technology capability from scratch** in a board-level quality unit that ran on spreadsheets: ETL pipelines (Python/SQL, Azure Functions, Azure ML), KPI frameworks, and Power BI dashboards, bringing precision and governance to previously unreliable metrics.
- Ran board-level studies: **academic dropout probability** and **student talent mobility** (entrance exam through graduation) using probabilistic modeling and graph maps; presented directly to directors, turning analysis into strategic decisions.
- Packaged the platform in **Docker** (multi-stage; docker-compose with API, UI, Qdrant and PostgreSQL with healthchecks) and used **Kubernetes** self-healing on a separate service.

Senior Data Scientist - BINAHKI

Sep 2023 - Aug 2024 · Remote, Brazil

Stack: Python, SQL, AWS (S3, SageMaker, EC2, Lambda), PostgreSQL, TensorFlow, Keras, LSTM, ARIMAX, XGBoost, NumPy, SciPy, computer vision (object classification), IoT

- Developed **anomaly detection** for compressors and condensers - whose failure always halted production - capturing weak early signals for preventive maintenance and reducing unplanned-downtime risk.
- On industrial furnaces, a high-accuracy **LSTM** over non-invasive electrical sensors exposed a **methodological flaw** responsible for significant, recurring energy waste.
- Hosted solutions in **production on AWS** (SageMaker, S3, EC2) with a PostgreSQL database, serving models online.
- Built energy-consumption **forecasting** (monthly, ARIMAX) supporting operational decisions, and **computer vision** (object classification).
- Designed an end-to-end AWS data architecture that underpinned **award-winning innovation proposals**.

Researcher (Ph.D.) - CBPF (Brazilian Center for Physics Research)

Nov 2017 - Jan 2024 · Rio de Janeiro, Brazil

- Independent research with high-impact publications; honorable mention for best thesis (2023) and FAPERJ Nota 10 scholarship; visiting researcher at the University of Warwick (UK).
 - Modeled large volumes of complex experimental data (structural, electrical and magnetic properties) with **advanced statistics**, extracting patterns and relationships from noisy data.
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Selected Projects

Agentic Curation System (RAG, LLMs) - End-to-end AI agents + RAG with structured output (Pydantic), human-in-the-loop, and an eval harness (LLM-as-judge, cost/latency/quality) against a baseline; FAISS in POC, Qdrant as the definitive vector DB (index, persistence, metadata, scale); vendor-agnostic client (Claude/Ollama); portable to LangChain/LangGraph. *Python, FastAPI, Qdrant, SentenceTransformers*.

Fraud Detection (KYC/KYT) - Supervised vs. unsupervised modeling to flag illicit transactions; XGBoost at high Average Precision, detecting most fraudulent activity. *Python, Scikit-learn, XGBoost, TensorFlow*.

Logistic Data Assessment - Exploratory and predictive analytics for operational efficiency: delay-risk prediction via gradient boosting with **SHAP** explainability, KPI monitoring, and dashboards; a significant reduction in average departure delay. *Python, Pandas, Polars, Scikit-learn, LightGBM, Plotly, FastAPI, SHAP*.

Voltage Time Series (LSTM) - Recurrent neural network for voltage forecasting and anomaly detection in electrical systems, with a full time-series preprocessing pipeline. *Python, TensorFlow, Keras, NumPy, Pandas*.

Interactive Analytics Apps - Web apps for data visualization and analysis. *Python, Streamlit, Pandas, Plotly, Scikit-learn*.

Magnetism Techniques (Thesis) - Research and analysis code for magnetism characterization techniques. *Python*. (github.com/Calazansmacchiutti/Magnetism-techniques-thesis)

Experimental Physics & Research Competencies

- **Materials:** synthesis of mono- and polycrystals (sol-gel, co-precipitation, floating zone); optimization of structures and **exchange-bias** effects in double perovskites.
 - **Characterization:** XPS, Raman spectroscopy, AFM, magnetometry; statistical analysis of complex experimental datasets.
 - **Research skills:** hypothesis design, controlled experiments, rigorous validation, scientific writing and peer-reviewed publishing.
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Additional Experience - Music & Digital Marketing

- Managed a band's digital presence and ran **Google Ads** campaigns optimized for **ROAS** (tickets, Spotify streams, followers), with conversion tracking and geo-targeting around show cities; roughly **doubled ROAS** and drove a **~10x increase in Spotify plays** on a tightly managed budget.
 - Hands-on with **YouTube and Instagram** content and audience-growth (algorithm/engagement) as a musician and producer.
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Education

- **Ph.D. in Physics** - CBPF (2020-2024) · Thesis: optimization of the exchange-bias effect in $\text{La}_{1.5}\text{Sr}_{0.5}\text{CoMnO}_6$
- **M.Sc. in Physics** - CBPF (2018-2020) · Synthesis and characterization of $\text{La}_{0.5}(\text{Ca},\text{Sr})_{0.5}\text{CoMnO}_6$ single crystals
- **B.Sc. in Physics** - UERJ (2012-2017)

Recognition & Research

- Publications in **Physical Review B** and **Physical Review Materials**
- Academic profiles: [Google Scholar](#) · [Lattes](#) · ORCID 0000-0002-8571-8876

Languages

Portuguese (native) · English (professional working proficiency)